

# Continuous Identities, Formalization and Explicit Budgets for Binary Goldbach

Synthesis V3.1 with Numerical Addendum

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## Abstract

This synthesis continues the V2 monograph while preserving its balance framework and notation. Research in round 22 transfers the additive coupling to continuous traces, Mellin integrals and character projections. The inventory frozen after independent batch 35 contains 88 modules and 1488 auxiliary declarations, including definitions. The new established results include exact extraction of the coefficient of the genuine von Mangoldt function on the circle, its alias-free discrete projection, canonical logarithm quantization at scale  $2^{58}$ , the five finite-field roots specified by the contract, and a Mellin representation with a justified interchange of sum and integral. The weighted truncation envelopes, their uniform geometry and their complete connection to the coefficient remain uncompiled proof sources. An exact rational test validates a scalar upper bound below  $10^{-6}$  at  $N = 10^8$ ; it does not evaluate the coefficient. A subsequent complete native computation at  $N = 10^8$ ,  $K = 2^{27}$  reconstructs the canonical integer  $I_N$  by five-modulus CRT and confirms equality with the direct  $A32$  sum and the auxiliary  $B40$  reconstruction. Their observed difference is zero, within the fixed quantization budget  $2E_{\log} < 10^{-6}$ . These observations do not establish native program refinement in Lean or validate the Mellin truncation chain. The formal inventory is unchanged in this numerical revision. The target bound for  $D_N$ , the global signed contribution and the binary Goldbach conjecture remain open.

**Keywords:** binary Goldbach; von Mangoldt; Mellin; continuous projection; explicit errors; Lean 4.

**Classification:** 11P32.

**Scope.** This text closes a research stage and provides a continuation contract. It announces neither a proof of Goldbach nor a breakthrough of the parity barrier. The statuses refer to preserved receipts, not to the number of identities that might still be formalized.

The manuscript was prepared with assistance from research agents and writing tools. This assistance does not constitute external scientific validation. Compiled proofs are identified by their sources and independent audits of their axioms.

# 1 Preserved framework and verification boundary

For an even integer  $N$ , we retain the notation

$$u = \log N, \quad L = 1 + u, \quad \ell = \log u,$$

with natural logarithms. The onset of the fixed analytic framework is

$$u \geq 10^{24}. \quad (1)$$

The point  $N = 10^8$  serves as a finite testing guide; it lies outside this regime. A check at this point covers neither the onset (1) nor all even integers below it. This stage preserves the established framework and examines its connection to new continuous objects.

The reference quantity in the balance remains

$$R_{\text{ref}} = \mathfrak{S}(N)(N - F_N), \quad \mathfrak{S}(N) = 2C_2 \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2}, \quad C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right). \quad (2)$$

The earlier analytic dependencies of  $F_N$  and the reference retain their status in the monograph. This synthesis does not turn a reported analytic input into a new self-contained Lean theorem. The research target remains

$$D_N \leq \tau_N := \frac{N}{256u\ell}. \quad (3)$$

The cutoffs of the preserved profile are

$$\alpha = \lceil N^{1/4} \rceil, \quad Q = \left\lfloor \frac{N-1}{\alpha} \right\rfloor, \quad a_{\text{cut}} = \lceil N^{7/16} \rceil, \quad M_{\text{cut}} = \lceil N^{3/4} \rceil.$$

The thermal parameter  $a > 0$  introduced below is distinct from  $a_{\text{cut}}$ . All boundary faces, bands, corners, units and support restrictions in the balance retain their positions.

## Four levels of evidence

| Level                | Exact meaning in this synthesis                                                                                         |
|----------------------|-------------------------------------------------------------------------------------------------------------------------|
| Compiled Lean        | Frozen source, successful independent compilation, exact catalogue and axiom audit; credit applies to the whole module. |
| SOURCE               | Written and possibly reviewed Lean proof text; elaboration and compilation are still absent or have failed.             |
| PAPER                | Identity or estimate derived in a note; its formalization and evaluation obligations remain explicit.                   |
| Observed computation | Program actually run under a precise contract; its result receives credit only within the scope of that contract.       |

The 1 488 counted declarations include definitions. They are not 1 488 Goldbach theorems. The standard axioms appearing in the audits include `propext`, `Classical.choice` and `Quot.sound`. No `sorryAx` is accepted in a credited module. Recovery output after failed elaboration remains evidence of a technical failure, with no proof credit.

## 2 The genuine thermal trace and the additive coefficient

We use the real arithmetic function

$$\Lambda(0) = \Lambda(1) = 0, \quad \Lambda(n) = \begin{cases} \log p, & n = p^k, \text{ } p \text{ prime, } k \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

All prime powers are retained. For  $a > 0$  and  $\theta \in \mathbb{R}$ ,

$$T_a(\theta) = \sum_{n \geq 0} \Lambda(n) e^{-an} e^{in\theta}, \quad C_N = \sum_{n=0}^N \Lambda(n) \Lambda(N-n). \quad (4)$$

No abstract trace function is needed: the series above is the object actually formalized.

Set  $r = e^{-a} \in (0, 1)$  and

$$U(a) = \frac{r}{(1-r)^2}, \quad (5)$$

$$B(a, M) = r^{M+1} \left( \frac{M+1}{1-r} + U(a) \right). \quad (6)$$

**Compiled Lean proof.** Batch 11 directly constructs  $0 \leq \Lambda(n) \leq n$ , summability of the terms, continuity of  $T_a$ , and the uniform bounds

$$|T_a(\theta)| \leq U(a), \quad |T_a(\theta) - T_{a,M}(\theta)| \leq B(a, M), \quad T_{a,M}(\theta) = \sum_{n=0}^M \Lambda(n) r^n e^{in\theta}.$$

It also proves continuity of the envelopes for  $a > 0$  and bounds the difference between the full and partial integral projections by  $2e^{aN}U(a)B(a, M)$ . The geometric majorant is constructed; it is not supplied as a precision hypothesis.

**Theorem 2.1** (Exact continuous projection; batch 13). *For every  $a > 0$  and every  $N \in \mathbb{N}$ ,*

$$C_N = \frac{e^{aN}}{2\pi} \int_0^{2\pi} T_a(\theta) \overline{T_a(-\theta)} e^{-iN\theta} d\theta = \frac{e^{aN}}{2\pi} \int_0^{2\pi} T_a(\theta)^2 e^{-iN\theta} d\theta. \quad (7)$$

**Compiled Lean proof.** The proof first extracts the characters on a finite rectangle, establishes integral orthogonality and the normalization factor, then passes to the limit using the envelope (6). Once  $M \geq N$ , the integral projection of the polynomial  $T_{a,M}$  already equals  $C_N$  exactly.

Conjugation applies to the reflected phase:  $\overline{T_a(-\theta)} = T_a(\theta)$ . The product  $|T_a(\theta)|^2$  would select frequency differences; it does not yield the required additive coefficient. The form centered on  $[-\pi, \pi]$  relies on periodicity of the genuine series. Its combination with the Mellin truncation is written separately, with the SOURCE status described in Section 8.

### 3 Alias-free finite projector and finite-field projection

The continuous identity has a fully finite counterpart. It provides an exact contract for the completed finite computation without replacing the genuine function  $\Lambda$ . For  $K \geq 1$ , set  $\theta_j = 2\pi j/K$ .

**Theorem 3.1** (Discrete orthogonality and extraction; batch 15). *For  $N \leq M$  and*

$$K > \max(N, 2M - N), \quad (8)$$

*for every real  $a$ , we have*

$$C_N = \frac{e^{aN}}{K} \sum_{j=0}^{K-1} T_{a,M}(\theta_j)^2 e^{-iN\theta_j}. \quad (9)$$

**Compiled Lean proof.** Batch 15 proves orthogonality of the actual exponential characters, then their expansion and selection of the antidiagonal. Condition (8) excludes spurious frequencies  $m + n - N \in K\mathbb{Z}$  with  $0 \leq m, n \leq M$ . It is a mathematical guard, not an experimental tolerance. The choice  $a = 0$  is valid for this finite polynomial; no infinite trace  $T_0$  is introduced.

An analogous projection exists in a finite field. For a prime  $p$ , an element  $\omega \in \mathbb{F}_p$  of exact order  $K$  and  $K \neq 0$  in  $\mathbb{F}_p$ , orthogonality gives

$$\frac{1}{K} \sum_{j=0}^{K-1} \omega^{jk} = \begin{cases} 1, & K \mid k, \\ 0, & K \nmid k. \end{cases}$$

**Compiled Lean proof.** Batch 19 proves this identity from the primitive root; the resulting orthogonality is not a premise. For a finite sequence  $A_n$  and (8), its projection selects  $\sum_{n=0}^N A_n A_{N-n}$  in  $\mathbb{F}_p$ .

The continuous-circle representation and this character projection share one object: the exact additive coefficient. A projection identity does not show that this coefficient is sufficiently large. Nor does it establish that a fast-transform program implements the theorem's finite sum. That correspondence requires stage invariants, normalizations, machine-word bounds and integer reconstruction.

The final contract fixes

$$N = M = 100\,000\,000, \quad K = 2^{27} = 134\,217\,728, \quad S = 2^{58} = 288\,230\,376\,151\,711\,744. \quad (10)$$

Here  $K > N$  satisfies (8). The finite proofs remain valid regardless of whether a computation of this size is completed. The validation contract must retain coefficients from all prime powers, including those of exponent at least two.

## 4 Canonical logarithms and explicitly constructed precision

The quantization in batch 17 does not assume oracle-supplied precision. For  $2 \leq p \leq 10^8$ , set

$$k = \lfloor \log_2 p \rfloor, \quad z = \frac{p - 2^k}{p + 2^k}, \quad 0 \leq z < \frac{1}{3},$$

and define the rational sum

$$L_{32}(z) = 2 \sum_{j=0}^{31} \frac{z^{2j+1}}{2j+1}, \quad \rho_{32} = \frac{9}{4 \cdot 65 \cdot 3^{65}}. \quad (11)$$

The real series representation of  $\log(1+z) - \log(1-z)$ , its geometric tail and the decomposition using  $2^k$  give the interval

$$b_p = kL_{32}(1/3) + L_{32}(z), \quad \log p \in [b_p, b_p + (k+1)\rho_{32}]. \quad (12)$$

**Compiled Lean proof.** On the fixed domain, the width satisfies

$$(k+1)\rho_{32} \leq \frac{243}{260 \cdot 3^{65}} < 2^{-96}.$$

The canonical point  $A_p$  is obtained by multiplying the midpoint of (12) by  $S$ , rounding to the nearest integer with ties to even, then clamping to  $[0, 32S]$ . The constructor uses exact rational quotient and remainder. It is not identified with a different rounding function without proof.

For all integers, define

$$A_n = \begin{cases} A_{\min \text{Fac}(n)}, & n \text{ is a prime power,} \\ 0, & \text{otherwise,} \end{cases} \quad q_n = A_n/S.$$

The cases 0 and 1 are handled separately. Using the direct definition of  $\Lambda$  and its least prime factor retains every  $p^k$ . **Compiled Lean proof.** The proof establishes, for  $0 \leq n \leq 10^8$ ,

$$0 \leq \Lambda(n), q_n \leq 32, \quad |\Lambda(n) - q_n| \leq 1/S. \quad (13)$$

Precision follows from (11)–(12) and the constructed rounding rule; it is not a free hypothesis of the theorem.

Set  $I_N = \sum_{n=0}^N A_n A_{N-n} \in \mathbb{Z}_{\geq 0}$ . The same batch proves

$$\left| C_N - \frac{I_N}{S^2} \right| \leq E_{\log}(N, S), \quad E_{\log}(N, s) = (N+1) \left( \frac{64}{s} + \frac{1}{s^2} \right), \quad s > 0. \quad (14)$$

The envelope is continuous for  $s > 0$ . The fixed integer guard

$$2(N+1)(64S+1)10^6 \leq S^2 \quad (15)$$

is compiled, hence  $2E_{\log}(10^8, 2^{58}) \leq 10^{-6}$ . This result proves the real quantization bound without computing  $I_N$  or  $C_N$ .

## 5 Five prime moduli, exact residues and native-code obligations

The contract fixes the following five pairs; their primality and the root orders were independently compiled in batch 22.

| $i$ | $p_i$         | $(p_i - 1)/K$ | $g_i$ |
|-----|---------------|---------------|-------|
| 1   | 2 013 265 921 | 15            | 31    |
| 2   | 2 281 701 377 | 17            | 3     |
| 3   | 3 221 225 473 | 24            | 5     |
| 4   | 3 489 660 929 | 26            | 3     |
| 5   | 3 892 314 113 | 29            | 3     |

In  $\mathbb{F}_{p_i}$ , set  $\omega_i = g_i^{(p_i-1)/K}$ . The proof establishes order  $K$  from primality certificates and the required power checks.

**Theorem 5.1** (Concrete canonical projection; batch 22). *With parameters (10) and the canonical integers  $A_n$ ,*

$$F_i(j) = \sum_{n=0}^N A_n \omega_i^{jn}, \quad r_i = K^{-1} \sum_{j=0}^{K-1} F_i(j)^2 \omega_i^{-Nj} = I_N \pmod{p_i}. \quad (16)$$

**Compiled Lean proof.** The theorem instantiates the finite-field projection with the actual roots in the table. Neither the final residue nor coefficient precision is taken as a premise.

**PAPER: written argument.** For  $\Pi = \prod_{i=1}^5 p_i$ , the contract bounds are

$$0 \leq I_N \leq (N+1)(32S)^2 < 2^{153} < \Pi. \quad (17)$$

They provide the domain for unique reconstruction by the Chinese remainder theorem. A reconstruction  $c \in [0, \Pi)$  with the five correct residues must therefore equal  $I_N$ , provided the exact connection to the constructor and projections is established. Proving the behavior of the CRT code, its machine words and the fast transform remains separate from (16).

The native producer and checker were both built in BUILD04: two compilations completed with exit code zero. This establishes the existence of two frozen executable images without proving their refinement of the Lean functions. The completed operational checker verifies the entire catalogue up to  $N$ , the classification of prime powers, equality of every recorded point to the canonical A32 constructor, residues, reconstruction and integer guards (Section 10). These are observed native checks, not a Lean program-refinement theorem. An auxiliary check using a B40 series is not the primary A32 constructor; its code and real precision have no complete Lean proof at this stage.

The resource window is an operational constraint. Declared trust in Windows, GCC and the native images is neither universal loader closure nor observation of all dynamic dependencies. None of these operational assumptions replaces a program-refinement theorem.

## 6 Mellin: exact identity with the genuine von Mangoldt series

Let  $w \in \mathbb{C}$  satisfy  $\Re w > 0$ . The principal logarithm determines  $w^{-s} = \exp(-s \operatorname{Log} w)$ . Set

$$s(t) = 2 + it, \quad D(t) = \sum_{n \geq 2} \Lambda(n) n^{-2-it}, \quad P(w) = \sum_{n \geq 0} \Lambda(n) e^{-nw}, \quad K(w, t) = \Gamma(2 + it) w^{-2-it}. \quad (18)$$

The index-zero and index-one terms of  $D$  vanish in the formal definition. For  $n \geq 1$ , the powers use the real logarithm of the positive number  $n$ . In particular,  $P(a - i\theta) = T_a(\theta)$ .

**Theorem 6.1** (Inversion and infinite interchange; batch 35). *For  $\Re w > 0$ , the function  $t \mapsto D(t)K(w, t)$  is integrable and*

$$P(w) = \frac{1}{2\pi} \int_{\mathbb{R}} D(t) \Gamma(2 + it) w^{-2-it} dt. \quad (19)$$

Moreover,  $D$  is continuous and  $|D(t)| \leq 6$  for every  $t \in \mathbb{R}$ .

**Compiled Lean proof.** Batch 35 contains 30 declarations: 24 theorems and 6 definitions. Its independent compilation and exact axiom audit succeeded. The interchange does not rely on an assumed final integrability statement or an externally supplied equality.

The proof begins with the positive series

$$b_n = \Lambda(n) n^{-2}, \quad 0 \leq b_n \leq 2n^{-3/2} \ (n \geq 1), \quad \sum_n b_n \leq 6.$$

It constructs uniform convergence of the series  $D$  and its bound. The Gamma inversion identities, holomorphy on the right half-plane and integrable kernel majorants come from independently compiled dependencies. For each  $n > 0$ , the branch identity  $(nw)^{-s} = n^{-s} w^{-s}$  is established with  $n$  a positive real number. Finally, summability of the integrals of norms justifies the actual interchange between the series in  $n$  and the integral in  $t$ . The terms  $\Lambda(p^k)$  remain present.

On the line  $\Re s = 2$ , the classical identification of this series with  $-\zeta'(s)/\zeta(s)$  belongs to a separate zeta connection. Theorem 6.1 directly uses the genuine series  $D$ ; its credit does not validate all explicit formulas in zeros, all contour shifts or a uniform evaluator for  $\zeta$ .

Identity (19) transfers the coupling exactly into a continuous representation. It does not yet give a favorable sign for coefficient (7). Absolute domination ensures existence of the integrals and the interchanges; a new signed cancellation remains a separate obligation.

## 7 Signed truncations and right-half-plane geometry

We retain the concrete definitions of the truncation packages. For  $\Re w > 0$ , let

$$\eta(w) = \frac{\pi/2 + |\operatorname{Arg} w|}{2}, \quad \delta(w) = \frac{\pi/2 - |\operatorname{Arg} w|}{2} > 0, \quad C(w) = |w|^{-2} \sec^2 \eta(w). \quad (20)$$

The Gamma kernel has constructed exponential majorants on this domain. Independent, compiled batch 30 establishes the unweighted tails in both orientations. The bridge in batch 32 connects full Gamma inversion to the corresponding truncation. These credits do not automatically include the new tail weighted by  $D$ .

**SOURCE: not compiled.** The  $E_\Lambda$  package defines, for  $H \geq 0$ ,

$$P_H(w) = \frac{1}{2\pi} \int_{-H}^H D(t) K(w, t) dt, \quad (21)$$

$$T_+(w, H) = \int_H^\infty D(t) K(w, t) dt, \quad T_-(w, H) = \int_H^\infty D(-t) K(w, -t) dt. \quad (22)$$

It aims to establish the exact statements

$$P(w) - P_H(w) = \frac{T_-(w, H) + T_+(w, H)}{2\pi}, \quad |P(w) - P_H(w)| \leq \frac{6C(w)e^{-\delta(w)H}}{\pi\delta(w)}. \quad (23)$$

The factor 6 comes from Theorem 6.1. Reflection applies to both factors  $D$  and  $K$ . The source contains concrete integrability and tail proofs; no premise equal to (23) is added.

Batch 36 fails on a reflection rewrite at line 118:6. It produced no compiled proof object for this module; the following geometry module was not invoked. The repair is preserved in a new SOURCE02 package, with no further compilation at the close of this stage. The diagnostic concerns elaboration, without an analytic refutation or a proof of a parity obstruction.

**SOURCE: not compiled.** The Geometry25 package constructs, for  $a > 0$ ,

$$C(a - i\theta) = \frac{2}{a^2} \left( 1 + \frac{|\theta|}{\sqrt{a^2 + \theta^2}} \right) \leq \frac{4}{a^2}. \quad (24)$$

For  $|\theta| \leq \pi$ , it also aims to establish

$$\delta(a - i\theta) \geq d(a) := \frac{1}{2} \arctan(a/\pi) > 0. \quad (25)$$

Branches, quadrants, signs, square roots and continuity are explicitly written out. The radii are closed expressions and continuous on their domains:  $\Re w > 0$  and  $H \in \mathbb{R}$  for the expression,  $H \geq 0$  for the tail bound. Continuity of an expression is not an additional proof of continuity of an integral with a moving boundary.

## 8 Coefficient envelope and remaining scalar connection

From (23)–(25), the proposed uniform radius is

$$\varepsilon(a, H) = \frac{24e^{-d(a)H}}{\pi a^2 d(a)}, \quad a > 0, \quad H \geq 0. \quad (26)$$

The coefficient source defines the centered projection of the truncation

$$C_{N,H} = \frac{e^{aN}}{2\pi} \int_{-\pi}^{\pi} P_H(a - i\theta)^2 e^{-iN\theta} d\theta$$

and the closed-form radius

$$\mathcal{E}_N(a, H) = e^{aN} (2U(a)\varepsilon(a, H) + \varepsilon(a, H)^2). \quad (27)$$

**SOURCE: not compiled.** The 34 declarations in this package construct transport to the centered circle, angular continuity and integrability at fixed  $H$ , then aim to establish

$$|C_N - C_{N,H}| \leq \mathcal{E}_N(a, H). \quad (28)$$

The elementary inequality used is  $|x^2 - y^2| \leq 2|x||x - y| + |x - y|^2$ . Together with  $|T_a| \leq U(a)$ , it explains the two budget terms. Joint continuity of the radius is written for  $a > 0$  and  $H \in \mathbb{R}$ ; its interpretation as a truncation error requires  $H \geq 0$ . The theorem is not compiled at stage closure.

The genuine series  $T_a$  is periodic. The complex power  $(a - i\theta)^{-2-it}$ , taken alone on its principal branch, is not. The source does not postulate periodicity of  $P_H$ . Shifting the projection interval concerns the full trace, whose periodicity follows from the series; the truncation then receives its own budget.

**SOURCE: not compiled.** The scalar package of 26 declarations defines the same functions  $U$ ,  $d$ ,  $\varepsilon$  and  $\mathcal{E}_N$ , and writes a real-variable chain at the point

$$N = 10^8, \quad a = 10^{-8} = 1/N, \quad H = 10^{11}, \quad \tau = 10^{-6}. \quad (29)$$

It aims in particular to prove  $d(a) > 1/(8N)$ , a bound for  $U$  through the sinh identity,  $e^{125} > 10^{50}$ , then  $\mathcal{E}_N(a, H) \leq 576/10^{10} < \tau$ . It is not compiled. A separate bridge of eight theorems writes the equalities between the two radius definitions and the instantiation of (28) by this scalar result. Its two imports are prospective; the bridge is also uncompiled.

These sources address precise theoretical obligations without establishing a numerical coefficient. Before their independent compilation, no complete Lean chain supports promoting (28) and its instantiation at (29) to an established formal result.

## 9 The completed rational test and its scope

The observed scalar test uses only exact fractions. At (29), it defines

$$S_{150}(125) = \sum_{k=0}^{150} \frac{125^k}{k!}, \quad q = \frac{1}{S_{150}(125)}, \quad \mathcal{B} = 3(128N^5q + 4096N^6q^2).$$

**Observed computation: rational PASS.** One actual invocation, with no retry, verifies exactly

$$\mathcal{B} < 10^{-6}. \quad (30)$$

The preserved checks include  $S_6(5) > 100$  and  $S_{150}(125) \geq S_6(5)^{25} > 10^{50}$ . The result is linked to the executor's receipt and the ROOT physical observation listed in the appendix. It uses no floating-point approximation.

Comparison (30) is a scalar result. The program does not integrate  $D$ , compute  $P_H$  or  $C_{N,H}$ , or evaluate  $C_N$ . It does not verify a prime catalogue, the native transform or a Goldbach identity. The connection between  $\mathcal{B}$  and the real radius (27) uses the analytic inequalities written in the scalar package and the bridge of Section 8. Their status remains SOURCE.

**PAPER: written argument.** The analytic connection to the same radius is explicitly as follows. The real inequalities in the connectors give  $3 < \pi < 7/2$ ,  $d(1/N) > 1/(8N)$  and  $U(1/N) = 1/(4 \sinh^2(1/(2N))) \leq N^2$ . Thus  $dH > 125$  at the fixed point. The positive Taylor series gives  $S_{150}(125) \leq e^{125}$ , hence  $e^{-dH} \leq e^{-125} \leq q$ . Expression (26) therefore satisfies

$$\varepsilon(1/N, H) \leq \frac{24}{3} N^2 (8N) q = 64N^3 q.$$

Since  $aN = 1$  and  $e < 3$ , (27) implies

$$\mathcal{E}_N(1/N, H) \leq 3(2N^2 \cdot 64N^3 q + (64N^3 q)^2) = \mathcal{B} < \tau.$$

This chain explains the proposed analytic scope of the test exactly. It remains PAPER/SOURCE until compilation of the connectors and the bridge to the coefficient; the rational computation alone does not verify these dependencies.

### Feasibility of a literal Mellin evaluation

**PAPER: written argument.** The sharper geometry gives the uniform radius

$$\overline{C}(a) = \frac{2}{a^2} \left( 1 + \frac{\pi}{\sqrt{a^2 + \pi^2}} \right) < \frac{4}{a^2}, \quad \overline{\varepsilon} = \frac{6\overline{C}(a)e^{-d(a)H}}{\pi d(a)}.$$

The cost note studies the threshold of the majorant  $e^{aN}(2U\overline{\varepsilon} + \overline{\varepsilon}^2)$ . For parameters (29), it places this threshold between  $5 \cdot 10^{10}$  and  $10^{11}$  using closed-form inequalities. This estimate concerns a sufficient condition obtained from a majorant; it is not a lower bound for the actual error and does not impose this height on every possible method.

The same note bounds a literal tail of  $D$  on paper by  $(\log M + 1)/M$  and angular quadrature through a derivative majorant. Extremely crude choices such as  $M = 10^{52}$  and  $J = 10^{67}$  illustrate symbolic closure of these budgets. They describe neither an available program nor a run that could finish within this stage's resource window. Directed certification of the primitives at every node and vertical quadrature remain open. The scalar success (30) is therefore not a performance measurement of a global Mellin evaluation.

## 10 Completed native evaluation and its verification boundary

The finite contract in Section 5 gives a falsifiable target: produce  $I_N$  from the canonical coefficients, reconstruct the same integer from the five residues, and verify all guards. Then (14) provides the interval

$$C_N \in \left[ I_N/S^2 - E_{\log}, I_N/S^2 + E_{\log} \right].$$

The radius is not chosen after observation; its real construction and guard are already compiled. The completed native checker verifies correspondence of the records to *A32*, complete catalogue coverage and the integer computation. Refinement of its machine implementation to the Lean objects remains a separate obligation.

**PAPER: written argument.** For  $K = 2^{27}$ , a radix-2 transform has  $K \cdot 27/2 = 1\,811\,939\,328$  butterflies per transform. The contract counts give tens of billions of modular products for the five moduli and their independent check. An array of  $K$  32-bit words occupies 536 870 912 bytes; the 64-bit points on  $0, \dots, N$  occupy 800 000 008 bytes, and the 32-bit factors occupy 400 000 004 bytes. These payload counts are neither a total memory measurement nor a runtime promise.

**Historical V3 observation: STOP without a verdict.** Consumer05 reused three frozen producer04 payloads as identical copies and a checker image from BUILD04. The payloads were not validated by that stopped attempt. The attempt creates one checker, no producer and no compiler, with no retry. The parent budget is 3 600 seconds; the checker has a deadline at 3 300 seconds, leaving a 300-second reserve for logs and preservation checks.

The checker stops with code 238 and status `HARD_WALL_NO_VERDICT`. The parent exits with code 1 after approximately 3 312 seconds, or 55 minutes. No coefficient was present in that receipt. Closure documents quiescence and the POST actually produced; the ROOT observation verifies preservation of 6 576 inputs, 109 capture pairs, three payload pairs and 3 089 archive files. The stop is a resource failure without a mathematical verdict, not a counterexample to the identity or the constructed precision.

Those OS resource controls distinguish a 2 GiB Job commit-memory limit from RSS monitoring, which is not an OS-enforced RSS limit. Operational trust in Windows/GCC/native code remains declared and bounded in scope; universal loader closure is not asserted. No new attempt was included in the V3 closure. These historical receipts remain unchanged.

**New observed computation: exact numerical arithmetic validated.** On 4 October 2026, a fresh execution of the unchanged BUILD04 producer and the operational checker both finish with exit code zero. The checker retains the reference *A32* construction and uses an exact integer Horner evaluation of its rational logarithm series, with a cached rational series for  $\log 2$ . A 128-case comparison of the exact lower and upper bounds against the reference implementation passes before the full run. The catalogue, prime-power classification, DIF transforms, CRT and direct integer sum remain within the same arithmetic contract.

The checker validates 5 761 455 prime records, all canonical words and the five residues. CRT reconstruction, the direct *A32* sum and auxiliary *B40* reconstruction agree exactly. The fresh producer’s three payload files are byte-identical to the checked copies. Appendix B records the integer, error radius, resource measurements and receipt bindings. The independent Python audit reconstructs CRT and checks receipts and hashes; it does not independently replay the full prime catalogue or transforms.

The operational Windows backend avoids privileged working-set quotas, the source of error 1314, and requests a 3 GiB Job commit-memory limit. RSS is monitored, not enforced as an OS working-set quota. The new checker and parent have a 24-hour wall budget; the unchanged producer finishes within its original internal deadline. An initial startup stop due to the console helper exceeding a one-process monitor is preserved; the completed invocations permit 16 active and 32 total Job processes. Neither completed launcher invocation performs an automatic retry.

## 11 Connection to the balance: the signed obligation remains

The coefficient  $C_N$  includes proper prime powers. Define directly

$$P_{\text{pr}}(n) = \begin{cases} \log n, & n \text{ prime,} \\ 0, & \text{otherwise,} \end{cases} \quad V_{\text{pp}}(n) = \Lambda(n) - P_{\text{pr}}(n).$$

The notation  $V_{\text{pp}}$  avoids confusing this weight with the amplitude  $U(a)$ . In the ledger audit, the coefficient of prime pairs and the correction are

$$R_{\text{sf}} = \sum_{n=0}^N P_{\text{pr}}(n)P_{\text{pr}}(N-n), \quad (31)$$

$$\begin{aligned} Q_N &= C_N - R_{\text{sf}} \\ &= \sum_{n=0}^N (2P_{\text{pr}}(n)V_{\text{pp}}(N-n) + V_{\text{pp}}(n)V_{\text{pp}}(N-n)) \geq 0. \end{aligned} \quad (32)$$

**PAPER: written argument.** The direct upper bound  $Q_N \leq 2\sqrt{N}u^2$  for  $N \geq 2$  is retained in this note. It is not credited here as a new Lean theorem. This global term is not interchangeable with the prime-power terms whose supports are separately retained in the balance.

The monograph's bridge, within its stated scope, under its conditions and with exactly the same support, retains

$$D_N + \varepsilon_{\text{tr}} = R_{\text{ref}} - R_{\text{sf}} + 2\max(e, 0). \quad (33)$$

Its connection to the continuous coefficient is therefore

$$D_N = R_{\text{ref}} - C_N + Q_N + 2\max(e, 0) - \varepsilon_{\text{tr}}. \quad (34)$$

The ledger retains each of  $B_{\text{prime}}^{(a_{\text{cut}})}$ ,  $B_{\text{pp}}^{(a_{\text{cut}})}$ ,  $P_{\text{band}, \geq 2}$ ,  $Z_{\text{face}, \geq 2}$ ,  $I_\alpha$  and  $2\max(e, 0)$  exactly once. This transposition does not redevelop these objects using the arithmetic methods abandoned in loop 22.

Knowing a narrow interval for  $C_N$  does not give the sign of  $R_{\text{ref}} - C_N$ , or control of  $e$ , the boundaries and the corners. To establish (3), a signed estimate compatible with (34) remains to be proved, retaining these obligations and the bridge conditions. Assuming this estimate as a premise equivalent to the target would not constitute a way around the parity barrier.

The continuous identities establish exact representations, and the budgets describe a truncation or quantization error. No global cancellation of the true spectral contribution transferred to the ledger has been proved. This obligation remains open after the present finite computation of  $I_N$  at  $10^8$  and the enclosure for  $C_N$  under the native correspondence assumptions. It must be addressed at the threshold (1) and with the finite coverage required by the framework.

## 12 Boundaries, zeros and invariance: directions with precise scope

The continuous pivot preserves phase before taking a norm. For  $a > 0$ ,  $N \geq 1$  and  $q \in \mathbb{C}$ , consider

$$J_{a,N}(q) = \frac{1}{2\pi} \int_{-\pi}^{\pi} (a - i\theta)^{-q} e^{-iN\theta} d\theta.$$

**Compiled Lean proof.** Batch 25 formalizes integration by parts and the true boundary term on the principal branch:

$$J_{a,N}(q) = \frac{i(-1)^N}{2\pi N} ((a - i\pi)^{-q} - (a + i\pi)^{-q}) + \frac{q}{N} J_{a,N}(q + 1). \quad (35)$$

The boundary term does not vanish pointwise. This module of 30 declarations provides no favorable sign for the full spectral projection. Periodicity of the trace after inversion does not make each truncated kernel periodic.

**PAPER: written argument.** The signed explicit-formula strategy localizes a smooth test function near  $x + y = N$ , then aims for the decomposition

$$C_N = \mathcal{B}_N - 2\mathcal{L}_N + \mathcal{Z}_N.$$

The three terms respectively denote the background, coupling to the zero channel, and the two-zero contribution in the proposed explicit formula. The full identity, archimedean corrections, multiplicities, interchanges and zero tails remain to be established in this channel. The retained reviews find consistency subject to these obligations; they credit neither a sign for  $\mathcal{Z}_N - 2\mathcal{L}_N$  nor a Lean proof of the global formula.

Organization into conjugation–reflection orbits retains zeros off the critical line. A four-point orbit generates a density of the form

$$4mx^{-1/2} \cosh((\beta - 1/2) \log x) \cos(\gamma \log x),$$

and a two-point critical orbit gives the density  $2mx^{-1/2} \cos(\gamma \log x)$ , with multiplicity  $m$ . Stabilizers prevent double counting a degenerate orbit. The cosine phase can change sign: correct organization of the orbits does not automatically give positivity of the additive coefficient.

The review of the canonical kernel exhibits a negative direction in a two-dimensional restriction; it rules out the naive positivity argument examined there. Its scope is this specific kernel. It is neither a universal impossibility result for every operator approach nor a refutation of Goldbach. Likewise, any proposed modularity or exact invariance must be proved for the true objects and must isolate the signed term required in (34). No such sufficient connection is presented as imminent at this closure.

### 13 Continuous Fejér: reconstruction and the remaining sign

The new spectral note starts from the true series  $D$  of batch 35, with  $c_n = \Lambda(n)/n^2$  for  $n \geq 2$  and  $c_0 = c_1 = 0$ . The mass bound  $\sum c_n \leq 6$  makes  $D$  positive definite: for finite families  $(z_j, t_j)$ ,

$$\sum_{j,k} \bar{z}_j z_k D(t_k - t_j) = \sum_{n \geq 2} c_n \left| \sum_k z_k e^{-it_k \log n} \right|^2 \geq 0.$$

**PAPER: written argument.** For  $T > 0$  and an integer  $m \geq 1$ , the triangular average

$$\begin{aligned} A_T(m) &= \frac{1}{T} \int_{-T}^T \left(1 - \frac{|t|}{T}\right) D(t) e^{it \log m} dt \\ &= \sum_{n \geq 2} c_n \operatorname{sinc}^2\left(\frac{T}{2} \log \frac{m}{n}\right) \end{aligned} \tag{36}$$

is real and nonnegative. Here  $\operatorname{sinc}(x) = \sin(x)/x$  if  $x \neq 0$  and  $\operatorname{sinc}(0) = 1$ . The interchange over the compact interval is justified by summability; the triangle is the autocorrelation of an interval indicator. Distinct frequencies satisfy  $|\log(m/n)| \geq \log(1 + 1/m)$ . The note therefore derives the closed radius

$$0 \leq A_T(m) - c_m \leq \frac{24}{T^2 \log^2(1 + 1/m)}, \quad T > 0. \tag{37}$$

For  $m = 1$ ,  $c_1 = 0$  and the same bound holds. The radius is continuous for  $T > 0$  and tends to zero for fixed  $m$ . Neither this identity nor its connecting results are compiled in this stage.

This mechanism reconstructs amplitudes from the true vertical data, then can insert them into the reflected blocks  $n, N - n$ . It does not create shared mass for a pair. The note retains a bounded countermodel: the sector consisting solely of powers of 2 is positive definite and satisfies the same reconstruction, while its additive coefficient at  $N = 14$  is zero. This observation rules out deducing a uniform strict lower bound from this positivity alone. It does not replace the true zeta function and does not rule out an additional identity specific to its full data.

A future evaluator would have to produce intervals for the true integrals  $A_T(m)$ , with controlled primitives, locations and quadratures. (37) does not account for any of these evaluation errors. The cost increases with  $m$ , and no feasibility at  $10^8$  is asserted. The earlier identities for divisible grids and their covariances remain in V2, with their support conventions; this direction concerns the continuous spectral variable  $t$ .

Research is frozen. Any resumption must establish an additional signed connection to the ledger (34), rather than rename this reconstruction as parity cancellation. No mechanism sufficient for (3) or imminent discovery is claimed.

## 14 Frozen formal inventory and record of limitations

The independent observation of batch 35 fixes the official inventory at **88 modules / 1 488 auxiliary declarations, including definitions**. Batch 36 adds no credit. The following table isolates the elements used in this synthesis; it is not an exhaustive total of the historical inventory.

| Batch | Object                                        | Declarations | Status                     |
|-------|-----------------------------------------------|--------------|----------------------------|
| 11    | Geometric envelope of the true thermal trace  | 42           | Row PASS ;<br>batch FAILED |
| 13    | Continuous extraction identity                | 20           | Lean PASS                  |
| 15    | Orthogonality and discrete complex projector  | 29           | Lean PASS                  |
| 17    | Canonical logarithms and envelope             | 52           | Lean PASS                  |
| 19    | Generic finite-field projector                | 24           | Lean PASS                  |
| 22    | Five concrete roots and $A_{32}$ projection   | 28           | Lean PASS                  |
| 25    | Integration by parts and boundary term        | 30           | Lean PASS                  |
| 26/27 | Local and holomorphic Gamma inversion         | 22/15        | Row PASS26 ;<br>PASS27     |
| 30/32 | Gamma tails and unweighted bridge             | 22/2         | Lean PASS                  |
| 35    | Mellin transform of the true $\Lambda$ series | 30           | Lean PASS                  |
| 36    | Weighted tail $E_\Lambda$ ; geometry          | 16 ; 25      | FAIL ; not in-<br>voked    |

Batches 11 and 26 have a global FAILED status: each retains one whole-module PASS row (Envelope42 and Local22, respectively), while another module in the same batch fails. Their partial credits are attested by the adjudications and observations; the failed modules are not counted as established. Batches 13 and 27 later validate distinct corrected sources.

The coefficient sources (34), scalar connecting results (26) and the coefficient–scalar bridge (8) have not been compiled. Their declaration counts are source catalogues; they do not add to 1 488. Earlier **SOURCE ONLY** comments in some files that subsequently became PASS remain in the historical captures; the current status comes from the adjudication and receipt, not from retroactive rewriting.

Correction batches have preserved their failures: batch 16 contains series and rational-cast connecting issues, repaired and compiled in batch 17; batches 31 and 33 contain instance or continuity issues, repaired in the source actually compiled in batch 35. Batch 36 fails on reflection of the product of two functions. These diagnostics are not proofs of the parity barrier. Conversely, an auxiliary PASS is not a victory over that obstruction.

Verification is a property of the exact file, exact catalogue and its dependencies. The checks retain earlier logs, captures, sources and archives; they distinguish full reading of a text, targeted reading, metadata inventory and byte hashing. The historical companion file `evidence_v3.json` specifies the V3 boundary. The new `evidence_v3_1.json` binds this revision to its completed numerical receipts without modifying the historical registry.

## A Compact provenance and resumption

All paths below are relative to the corpus root `Goldbach_Parity_20261002`. SHA-256 prefixes aid reading; complete hashes, sizes, statuses and reading scopes are retained in `synthesis_v3/evidence_v3.json`. Source paths do not replace execution receipts. The published corpus is available in the repository `goldbach-horizon`, with the V3 companion registry.

| Item               | Relative path                                                                                               | SHA prefix   |
|--------------------|-------------------------------------------------------------------------------------------------------------|--------------|
| Mellin PASS        | <code>round22/judge5/batch35/sources/ComplexGammaMellinLambda22.lean</code>                                 | 4a94415678e4 |
| Receipt 35         | <code>round22/judge5/batch35/batch35_attempt01/receipt.json</code>                                          | 3cffa331b392 |
| Observation 35     | <code>.arbor/sessions/parity/.coordinator/messages/round22_judge_batch35_closed_observation.json</code>     | dc5985976051 |
| Repaired tail      | <code>round22/role4/complex_gamma_mellin_lambda_tail_source02/ComplexGammaMellinLambdaTail22.lean</code>    | See JSON     |
| Geometry SOURCE    | <code>round22/role4/complex_gamma_circle_geometry_source01/ComplexGammaCircleGeometry22.lean</code>         | 2322915060c6 |
| Coefficient SOURCE | <code>round22/role4/lambda_circle_truncation_source01/LambdaCircleTruncationEnvelope22.lean</code>          | bf9b8257760a |
| Scalar SOURCE      | <code>round22/role4/scalar_radius_real_connectors_source01/ScalarRadiusRealConnectors22.lean</code>         | 6ba4b8d4cca0 |
| Bridge SOURCE      | <code>round22/role3/lambda_circle_scalar_bridge_source01/LambdaCircleScalarRadiusBridge22.lean</code>       | cfb0d2cfd238 |
| Rational point     | <code>round22/role6/mellin_circle_radius_point_source01/actual_radius_point01_attempt01/receipt.json</code> | d46e3ffeb6d0 |
| Ledger             | <code>round22/role4/continuous_dn_gap_paper01/dn_gap_audit22.md</code>                                      | See JSON     |

### Self-contained handoff contract

The historical document `synthesis_v3/HANDOFF_THEORY_V3.txt` is preserved. Its successor `HANDOFF_FEJER_V3_1.txt` records the completed finite integer computation, receipt locations and unchanged proof obligations. A future Fejér session can resume from the existing PAPER note in Section 13; reconstruction, certified evaluation and the signed connection to the ledger must remain distinct. The tail, geometry, coefficient and connecting sources still require independent compilation, followed by certification of the primitives and quadratures actually chosen. No new Fejér research or Lean theorem is introduced by this documentary revision. A signed bound on the ledger remains an open obligation.

## B Numerical addendum at $N = 10^8$

**Observed native arithmetic; 4 October 2026.** The finite run uses

$$N = 100\,000\,000, \quad K = 134\,217\,728 = 2^{27}, \quad S = 288\,230\,376\,151\,711\,744 = 2^{58}.$$

The exact reconstructed integer is

$$I_N = 14616354094337692277256326339388864358034646.$$

The five residues, in the modulus order of Section 5, are

$$(577099750, 626366149, 232952773, 2618309304, 2671748987).$$

An independent Python CRT reconstruction gives the same integer as the producer, the native direct  $A32$  sum and the auxiliary  $B40$  reconstruction:

$$I_{\text{CRT}} = I_{\text{direct},A32} = I_{B40} = I_N.$$

Its normalized value is

$$I_N/S^2 = 175937962.675324617193216257686209932693 \dots$$

This number is the normalized canonical integer, not an exact evaluation of the real logarithmic coefficient  $C_N$  or of a truncated Mellin integral. The fixed primary radius from the canonical logarithm contract is

$$E_{\log} = \frac{1844674425817699235409551617}{83076749736557242056487941267521536} \approx 2.2204460714547736 \times 10^{-8}.$$

The observed normalized difference is zero, and exact rational comparison confirms

$$0 = |I_{A32} - I_{B40}|/S^2 \leq 2E_{\log} \approx 4.4408921429095471 \times 10^{-8} < 10^{-6}.$$

The  $C_N$  enclosure inherits the canonical correspondence assumptions; the auxiliary  $B40$  real-log refinement remains unproved in Lean.

| Process             | Wall (s) | CPU (s)    | Peak RSS (bytes) |
|---------------------|----------|------------|------------------|
| Producer BUILD04    | 1098.219 | 1097.65625 | 1743966208       |
| Operational checker | 1659.281 | 1657.84375 | 1744252928       |

The corresponding parent wall times are 1099.156 and 1661.266 seconds. The overlapping executions and finalization take 1678.252969 seconds (27 minutes 58.253 seconds). Both receipts confirm driver exit, an empty Job and validated POST observations. The peak RSS is about 1.625 GiB per native process; it is distinct from the requested 3 GiB commit quota.

**Unchanged boundary.** There are still 88 credited Lean modules and 1 488 auxiliary declarations, including definitions. Native Lean refinement, the complete Mellin coefficient truncation chain, spectral  $H1$ , control of  $D_N$  and Goldbach remain unproved. No new theory was undertaken for this run.

## New numerical provenance

The following paths are relative to `Goldbach_Numerical_20261004`. Full SHA-256 values, file sizes, exact fractions and execution metadata are retained in `numerical_verdict.json` and `evidence_v3_1.json`. The verdict string of the native checker is `EXACT_INTEGER_PROJECTION_CHECKED_PENDING_PRIMITIVE_PROOF`. It is not a proof-success flag for the open analytic obligations.

| Item             | Relative path                              | SHA prefix                |
|------------------|--------------------------------------------|---------------------------|
| Producer receipt | <code>producer_run02/receipt.json</code>   | <code>a9dfea2ffde7</code> |
| Checker receipt  | <code>checker_run01/receipt.json</code>    | <code>5f435444318f</code> |
| Build receipt    | <code>checker_build_receipt.json</code>    | <code>b647aa25994f</code> |
| Checker source   | <code>checker_dif22_operational.cpp</code> | <code>589f61d71717</code> |
| Checker image    | <code>checker_dif22_operational.exe</code> | <code>268c4a481691</code> |
| Factor payload   | <code>payload_run02/factors.bin</code>     | <code>32450e19fb3d</code> |
| Prime records    | <code>payload_run02/records.bin</code>     | <code>bf6bfeb2d843</code> |
| Producer claim   | <code>payload_run02/producer.txt</code>    | <code>aeee44e0e6cd</code> |

The new numerical archive preserves the Windows backend, checker, scripts, contracts, logs and receipts, including the initial infrastructure failure. The unchanged producer image remains in the historical V3 corpus, with its hash bound by the execution receipt.

The repository `goldbach-horizon` separates the original V3 publication from the new delivery and numerical archive:

```
publications/2026-10-goldbach-synthesis-v3-1
research/goldbach-numerical-20261004
```

The manifest and restore helper recover all original bytes and SHA-256 values, including losslessly fragmented large payloads. No file is omitted.

The entry point `HANDOFF_FEJER_V3_1.txt` accompanies this synthesis and its registry. Fejér retains PAPER status; this addendum closes the native run without starting that future research session.

## Local documentary sources

The following local sources preserve the written arguments and their limitations. No new external references or secondary citations were verified.

## References

- [1] Durand Serge, *Finite Identities and Explicit Error Budgets in a Framework for Binary Goldbach*, V2 synthesis, 1 October 2026. Title, author and continuity recorded from the retained public source.
- [2] Durand Serge, final monograph and local extraction `monographie.txt`; conventions updated by the source framework `Goldbach_Final_20261001/goldbach_synthesis.tex`.
- [3] ROLE4, *Finite projection contract*, `round22/role4/circle_ntt_paper01/projection_contract22_final.md`.
- [4] ROLE4, *Mellin-Lambda feasibility*, `round22/role4/mellin_lambda_feasibility_paper01/feasibility22.md`.
- [5] ROLE4, *Audit of the spectral connection to  $D_N$* , `round22/role4/continuous_dn_gap_paper01/dn_gap_audit22.md`.
- [6] Independent Judge, *Signed explicit-formula review*, `round22/judge5/signed_weil_source_review01/review22.md`.
- [7] ROLE4, *Zero-orbit review*, `round22/role4/zero_orbit_source_review01/review22.md`.
- [8] ROLE1, *True positivity of  $D$  and reconstruction of the retained blocks*, `round22/role1/spectral_next_paper01/spectral_next22.md`.